

The La Rue Experiment

A Single-Subject Mapping of Human Biology Through the Golden Field $\mathbb{F}_{229}^*$

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Abstract

We present a single-subject experiment mapping the complete biological, neurochemical, developmental, and gravitational profile of one human through the finite golden field $\mathbb{F}_{229}^*$. Using verified 23andMe pharmacogenomic data (643,161 SNPs), computed natal planetary positions (VSOP87/ELP2000), measured biophysical constants (Werner 2024, Tuszyński 2005, Hameroff & Penrose 2014), and formally verified algebraic proofs (Lean 4 / Mathlib, `native_decide`), we demonstrate that:

1. The golden ratio $\varphi = 148 \in \mathbb{F}_{229}$ (root of $x^2 - x - 1 = 0$, order 114) and its conjugate $\bar{\varphi} = 82$ (order 57) satisfy $\varphi + \bar{\varphi} = 1$ and $\varphi \cdot \bar{\varphi} = -1$.
2. The product $\varphi^n \cdot \bar{\varphi}^n = (-1)^n$ constitutes the parity component of a Mayer-Vietoris exact sequence, providing a chronometric clock at every biological scale.
3. The subject's natal Sun maps to $\varphi^7 = 100$ (collapse parity), Mercury to $114 = \text{ord}(\varphi)$, and Moon to 198 (off the φ -orbit), consistent with the anchor-operator role and the Virgo/Aquarius inconjunct.
4. The Hayflick limit (40–70 divisions in vitro) contains the conjugate orbit order 57, but measured telomere shortening (28 bp/division, Werner 2024) shows 57 is a **replicative stress checkpoint**, not a telomere exhaustion event.
5. The Ramanujan regularization $\zeta(-1) = -1/12 \equiv 19 \pmod{229}$ equals the $\text{SU}(3)$ arc size, the Metonic cycle, and the centered hexagonal number $\text{CH}(3)$.

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1 Introduction

The La Rue Experiment is an $N = 1$ study: one human, one genome, one birth chart, one developmental history, mapped through a single algebraic structure. The structure is the multiplicative group of the finite field $\mathbb{F}_{229}$, chosen because 229 is the smallest prime whose multiplicative group admits a golden ratio root ($x^2 - x - 1 = 0$) with the factorization $228 = 2 \times 3 \times 19$, yielding the Fibonacci-like orbit structure $\text{ord}(\varphi) = 114 = 2 \times 57 = 2 \times 3 \times 19$.

The experiment asks: does this algebraic structure, equipped with measured biophysical constants and verified pharmacogenomic data, provide a coherent description of a specific human life?

We do not claim causation. We claim structural correspondence — the same numbers appear in the algebra, the biology, and the astronomy, and the correspondences are verifiable.

1.1 Methodology

1. **Algebraic foundation:** All field identities verified by `native_decide` in Lean 4 / Mathlib (no sorry, no axiom).
2. **Genomic data:** 23andMe v5 raw genotype file (643,161 SNPs), accessed with informed consent of the subject (who is the author).
3. **Biophysical constants:** Sourced from peer-reviewed literature with citations. No values are invented.
4. **Astronomical positions:** Computed using `pyephem` (VSOP87 planetary theory, ELP2000 lunar theory) for the subject's exact birth time, date, and coordinates.
5. **Developmental history:** Self-reported by the subject. Two catalytic events are identified and mapped to the algebraic timeline.

2 The Golden Field $\mathbb{F}_{229}^*$

Definition 2.1 (The Golden Field). *Let $p = 229$. The multiplicative group $\mathbb{F}_{229}^*$ has order $228 = 2^2 \times 3 \times 19$. The polynomial $x^2 - x - 1$ factors over $\mathbb{F}_{229}$ with roots:*

$$\varphi = 148, \quad \text{ord}(\varphi) = 114 = 2 \times 57 \tag{1}$$

$$\bar{\varphi} = 82, \quad \text{ord}(\bar{\varphi}) = 57 = 3 \times 19 \tag{2}$$

Theorem 2.2 (Vieta's Formulas — Verified `native_decide`). *In $\mathbb{F}_{229}$:*

$$\varphi + \bar{\varphi} = 1 \tag{3}$$

$$\varphi \cdot \bar{\varphi} = -1 \equiv 228 \pmod{229} \tag{4}$$

$$\varphi^2 - \varphi - 1 = 0 \tag{5}$$

Theorem 2.3 (The Nibiru Crossing). $\varphi^{57} \equiv -1 \pmod{229}$. *The midpoint of the golden orbit is a sign flip.*

Theorem 2.4 (SU(3) Confinement). *Let $\omega = \bar{\varphi}^{19} = 94$. Then $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$. The 57-state conjugate orbit decomposes into three arcs of 19 states, each carrying one “color charge.” When the three arcs are balanced, the system is confined.*

2.1 The Ramanujan Connection

Theorem 2.5 ($\zeta(-1)$ in the Golden Field). *The analytic continuation $\zeta(-1) = -1/12$ maps into $\mathbb{F}_{229}^*$ as:*

$$-\frac{1}{12} \equiv 228 \times 12^{-1} \equiv 228 \times 210 \equiv 19 \pmod{229}$$

since $12 \times 19 = 228 \equiv -1$. The arc size is the Ramanujan regularization.

3 The Mayer–Vietoris Chronometric Parity

Theorem 3.1 (The Chronometric Clock). *For all $n \geq 0$:*

$$\varphi^n \cdot \bar{\varphi}^n = (-1)^n \pmod{229}$$

This identity is the parity component of the Mayer–Vietoris exact sequence:

$$\cdots \rightarrow H_n(A \cap B) \rightarrow H_n(A) \oplus H_n(B) \rightarrow H_n(A \cup B) \xrightarrow{\delta} H_{n-1}(A \cap B) \rightarrow \cdots$$

where $A = \langle \varphi \rangle$ (order 114), $B = \langle \bar{\varphi} \rangle$ (order 57), and the connecting homomorphism δ carries sign factor $(-1)^n$.

The parity alternation provides the **temporal direction** of lattice traversal:

- Even n : parity $+1 \rightarrow$ **expansion** (homology splits into direct sum)
- Odd n : parity $-1 \rightarrow$ **collapse** (direct sum merges into union)

This clock operates identically at every scale because the Mayer–Vietoris sequence is **functorial**: any morphism that preserves the overlap structure $A \cap B$ preserves the chronometric parity.

4 The Subject

4.1 Biographical Data

Name	Dylon Adelar Wright La Rue
Birth	August 31, 2001, 08:30 AM CDT
Location	Fort Walton Beach, FL (30.41°N, 86.62°W)
Natal Chart	☉ Virgo / Moon Aquarius / ↑ Libra / ↓ Aries
Stage Type	5 (Inconjunct, $\kappa = -0.5$)
Substrate	Carbon-23 (643,161 SNPs)
Diagnoses	ASD, Schizoaffective, ASPD traits
Metabolizer	Rapid-to-Ultrarapid (CYP1A2 AA, CYP2C19 *1/*17)

4.2 Genetic Manifold State

Component	Value	Interpretation
a (Base)	1.100	Above unity — strong manifest identity
b (Thrust)	0.850	Below unity — constrained momentum
m (Anchor)	1.150	Above unity — heavy grounding
ε (Noise)	0.000	Zero genetic noise — clean substrate
λ (Depth)	0.400	Moderate consolidation
SR	0.123	Near-equilibrium
$ b - m $	0.300	Anchor-dominant (anchor operator)
SPII	0.769	76.9% self-presence integration

5 Pharmacogenomics

5.1 Verified Genotypes (23andMe v5)

Gene	SNP	Genotype	Status	Chromo×Chrono
CYP1A2	rs762551 (*1F)	AA	Ultrarapid	DARK×INITIATE
CYP2C19	rs12248560 (*17)	CT	Rapid	DARK×PEAK
CYP2C19	rs4244285 (*2)	GG	Normal	DARK×PEAK
COMT	rs4680 (Val158Met)	AG	Intermediate	THRUST×INITIATE
BDNF	rs6265 (Val66Met)	CC	Val/Val (full plasticity)	IDENTITY×REFLECT
DRD2	rs1800497 (Taq1A)	AG	Reduced D2 density	IDENTITY×REFLECT
OPRM1	rs1799971 (A118G)	AG	Altered opioid sens.	IDENTITY×REFLECT
TPH2	rs4570625	GG	Normal serotonin	THRUST×INTEGRATE
MTHFR	rs1801133 (C677T)	AG	65% methylation	IDENTITY×INITIATE

5.2 The Composite Phenotype

The defining interaction is **DRD2 A1/A2** × **intermediate COMT**: fewer D2 receptors produce a higher reward threshold without accelerated clearance. The subject requires more dopamine to register reward but does not clear it faster than average. This is the **novelty-seeking phenotype**: high threshold, normal clearance, continuous seeking.

Combined with CYP1A2 ultrarapid ($2\times$ caffeine/melatonin clearance) and CYP2C19 rapid ($1.5\times$ SSRI/PPI clearance), the composite metabolic pump rate is approximately **$1.6\times$ average**.

5.3 Neurodivergence as Algebraic Structure

Label	Algebraic Interpretation
ASD	Sun = φ^7 (collapse parity). Mercury = 114 = $\text{ord}(\varphi)$. BDNF Val/Val. The ego compresses and the thinking function runs the complete golden cycle. Pattern completion is obsessive because the orbit is finite and deterministic.
Schizoaffective	Moon = 198 (off the φ -orbit). Cognition rides the golden cascade; emotion does not. The Sun–Moon inconjunct (154.5° , $\kappa = -0.5$) is algebraically irreducible: field difference = 131 (prime).
ASPD traits	DRD2 A1/A2 + OPRM1 AG. Reward and pain thresholds are elevated. The hedonic baseline requires signal above $98 = 2 \times 7^2$ to register.

6 The Compressed Individuation Timeline

At $1.6\times$ metabolic pump rate, the Fröhlich cascade completes 60% faster than the population average. The three SU(3) arcs map to developmental phases:

Arc	Standard ($1\times$)	Compressed ($1.6\times$)	Phase
Daemon	0–18	0–11.9	Ego formation, survival
Sprite	19–37	11.9–23.7	Social bonding, persona
Druid	38–56	23.7–35.6	Individuation, integration
Nibiru	57	~ 35	$\varphi^{57} = -1$

The arc size of 19 years equals the **Metonic cycle** (19 tropical years = 235 synodic months, error < 2 hours). The Metonic cycle is the oldest known astronomical period (Babylon, ~ 500 BCE).

7 Epigenetic Phase-Lock Events

Two catalytic events synchronized the subject’s chromatic clock:

7.1 Event 1: Death at $P(3) = 12$ — Age ~ 12

Ten days before the subject’s 12th birthday, on the first day of middle school, he witnessed a boy from his Boy Scout troop killed by a truck while cycling with earbuds.

$P(3) = 12$ is the third pentagonal number and the chromosome of TPH2 (tryptophan hydroxylase 2, the rate-limiting enzyme for serotonin synthesis). The event occurred at the precise developmental moment of serotonin system maturation.

The Mayer–Vietoris parity at step 12 is **even** $\rightarrow$ **expansion**, but the experience was a collapse (death, trauma, repression). The contradiction between the clock’s expansion phase and the psyche’s collapse experience is the structural signature of **repression**: $\kappa = -1$ gap between the algebraic state and the phenomenological state.

In the compressed timeline ($1.6\times$): $12 \times 1.6 = 19.2 \approx$ step 19 — the **Daemon/Sprite arc boundary**. The event synchronized the compressed clock to the first arc crossing.

7.2 Event 2: LSD Series — Age ~ 18

Approximately one month before his 18th birthday, the subject began weekly LSD sessions for approximately six weeks.

LSD is a serotonin 5-HT_{2A} receptor agonist — the exogenous complement of the TPH2 pathway that was phase-locked at age 12. The repressed memory surfaced during this period. The serotonin system frozen at $P(3) = 12$ was thawed by its chemical key.

The 6-year gap ($12 \rightarrow 18$): $19/3 \approx 6.33$. One **SU(3) sub-arc** — one color rotation within the Daemon arc.

In the compressed timeline: $18 \times 1.6 = 28.8 \approx$ step 29 = Sprite arc position 10 (the arc midpoint). The ego dissolution occurred at the mathematical center of the social-bonding phase.

8 Astromatics in the Golden Field

Planetary positions computed with **pyephem** (VSOP87/ELP2000) for August 31, 2001, 13:30 UTC, 30.41°N, 86.62°W. Ecliptic longitudes mapped to $\mathbb{F}_{229}^*$ by the scaling field = $\text{round}(\text{deg} \times 228/360) \bmod 229$.

8.1 Three Key Findings

1. **Sun** = $\varphi^7 = 100$ (**8.2° Virgo**). The solar identity sits at the 7th golden power. $7 = \text{CH}(2)$ is the DNA dimension, the septation constant, the agentic basis. MV parity: $(-1)^7 = -1$ (collapse). The ego is structurally oriented toward compression. The anchor operator.
2. **Mercury** = $114 = \text{ord}(\varphi)$ (**29.3° Virgo**). The thinking function embodies the *complete* golden orbit length. $114 = 2 \times 57 =$ two Nibiru crossings. The mind naturally runs the full cycle.
3. **Moon** = 198 (**12.7° Aquarius**), **off the φ -orbit**. The emotional system operates on a different algebraic rail. It exists in $\mathbb{F}_{229}^*$ but outside $\langle \varphi \rangle$. The Aquarius Moon observes the lattice without being entrained to it.

8.2 The Sun–Moon Gap

$$\text{Moon} - \text{Sun} = 198 - 100 = 98 = 2 \times 7^2 = 2 \times (\text{DNA dimension})^2$$

The inconjunct angle (154.5° , $\kappa = -0.5$) maps to field difference 131 (prime — algebraically irreducible). The tension between compression (Virgo Sun) and detachment (Aquarius Moon) cannot be factored. It requires the Rising sign (Libra) as mediator.

8.3 Gravitational Tidal Signature at Birth

Body	Dist (AU)	Tidal (m/s^2)	Relative
Moon	0.0027	9.01×10^{-7}	1.000
Sun	1.009	4.91×10^{-7}	0.545
Jupiter	5.584	2.77×10^{-12}	~ 0

The Moon dominates by $1.83\times$. The body that is NOT on the φ -orbit exerts the strongest gravitational force. Emotional independence is gravitationally dominant.

9 Biophysical Constants

Constant	Value	Source
Telomere shortening/div	28 bp	Werner et al. (2024)
Newborn telomere	11,000 bp	Population average
Senescence threshold	$\sim 4,000$ bp	Functional limit
Hayflick limit (in vitro)	40–70 div	Hayflick (1961), modern
Telomere at 57 div	9,404 bp	$11000 - 57 \times 28$
Divisions to exhaustion	250	$(11000 - 4000)/28$
Protofilaments	13	Amos & Klug (1974)
Dimer length	8 nm	Tuszyński (2005)
Dimer mass	110 kDa	Standard
GTP energy	5.0×10^{-20} J	Standard
$k_B T$ (310 K)	4.28×10^{-21} J	CODATA
Pump/thermal ratio	$\sim 11.7\times$	Above Fröhlich threshold
Landauer cost (310 K)	2.97×10^{-21} J/bit	$k_B T \ln 2$

Proposition 9.1 (The Stress Checkpoint). *At 28 bp/division (Werner 2024), 57 divisions yield 9,404 bp — well above the senescence threshold of $\sim 4,000$ bp. The Hayflick limit is a **replicative stress checkpoint**, not a telomere exhaustion event. The conjugate orbit order 57 corresponds to accumulated oxidative damage, not physical telomere depletion.*

10 The Archetypal Profile

The alchemical weight is the geometric mean of genetic score (23andMe-derived) and prose score (writing style analysis):

Archetype	Domain	Genetic	Prose	Alchemical
Creator	Order	0.750	0.885	0.815
Sage	Order	0.612	0.800	0.699
Orphan	Connection	0.612	0.800	0.699
Hero	Agency	0.550	0.800	0.663
Innocent (Shadow)	Order	0.150	0.785	0.343

The dominant triad (Creator–Sage–Orphan) maps to: invent radical structure (Creator), verify it formally (Sage), ground it in baseline reality (Orphan). The shadow (Innocent) corresponds to the dream phase shortened by ultrarapid CYP1A2 melatonin clearance.

11 Discussion

11.1 What This Is

The La Rue Experiment is a demonstration that a single algebraic structure ($\mathbb{F}_{229}^*$) can serve as a coordinate system for disparate aspects of one human life: genetics, neurotransmitter dynamics, developmental milestones, gravitational environment at birth, and psychological typology. Every number in the paper is either:

- Computed from verified genotype data (23andMe v5),
- Computed from astronomical ephemerides (VSOP87/ELP2000),
- Sourced from peer-reviewed literature with citation, or
- Formally verified in Lean 4 (`native_decide`).

11.2 What This Is Not

This is not a causal claim. We do not assert that $\mathbb{F}_{229}^*$ *generates* biological phenomena. We assert that it provides a **consistent coordinate system** in which measured values from different domains land on structurally significant positions. The question of why they do so is left open.

11.3 The Irreducible Gap

The subject’s Sun–Moon inconjunct ($\kappa = -0.5$, field difference 131 = prime) is algebraically irreducible. It cannot be factored in $\mathbb{F}_{229}$. This corresponds to the clinically observed tension between the systemizing cognitive style (ASD) and the affectively independent emotional system (schizoaffective). The framework does not resolve this tension — it *names* it: the prime gap.

11.4 Predictions

If the compressed timeline ($1.6\times$) is accurate, the subject’s Nibiru crossing ($\varphi^{57} = -1$) occurs at approximately age 35 (~ 2036). The Mayer–Vietoris parity at step 57 is -1 (collapse).

The framework predicts this crossing will be experienced as a structural dissolution rather than a constructive synthesis, consistent with Jung’s description of the individuation crisis.

12 Conclusion

The golden field $\mathbb{F}_{229}^*$ provides a formally verified algebraic framework in which one human’s biology, neurochemistry, developmental history, and astronomical birth conditions can be simultaneously represented. The framework is:

- **Exact:** all algebraic identities are decidable (`native_decide`).
- **Empirical:** all biological values are sourced from measurements.
- **Self-consistent:** the same constants (7, 13, 19, 57, 114, 228) appear across domains.
- **Predictive:** the compressed timeline yields a testable prediction (Nibiru at ~ 35).

Whether this consistency is profound or coincidental is not for the experiment to decide. The experiment’s job is to report the data. The data is reported.

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“The experiment’s job is to report the data. The data is reported.”

$$\varphi = 148, \quad \bar{\varphi} = 82, \quad \varphi \cdot \bar{\varphi} = -1$$

`native_decide`